

How to Test MPPT Controller Prototype

1. Cover the panel

Cardboard over it, or just disconnect. That's the off switch for everything below.

2. Flash

USB in, upload, open serial monitor at **115200 baud**.

3. Uncover the panel

4. Read the two boot lines

```
# panel meter (SDA 32 SCL 33 addr 0x40): OK  
# load meter (SDA 13 SCL 14 addr 0x40): OK
```

Both must say OK. Anything else, stop and debug/rewire.

5. Check Vin looks sane

Rows are already scrolling. The **Vin** column (4th) should read roughly **20–22 V** with duty at 0. If it's near zero or nonsense, stop.

6. Type **s** and press enter

The sweep runs about 30 seconds. Scan the **Pin** column (6th) and find the highest value.

Write down the duty number on that row. Probably around 2250.

7. Type **c** and press enter

Watch the **duty** column (3rd). It should climb toward the number you wrote down and then bounce up and down by 16 around it.

If it settles at the sweep's peak duty, the MPPT works.

8. Shade test

Still in **c**. Hold cardboard over about a third of the panel. Watch the duty column move to a new value. Uncover, watch it come back.